

What's happening now?

Spirit and *Opportunity* are two rovers that began exploring the Martian surface in 2004. Though *Spirit* stopped working in 2009, *Opportunity* is still operational. They have both sent back a wealth of data about the planet's surface, including evidence that huge areas of Mars were once covered by liquid water.

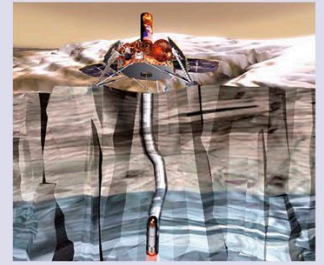
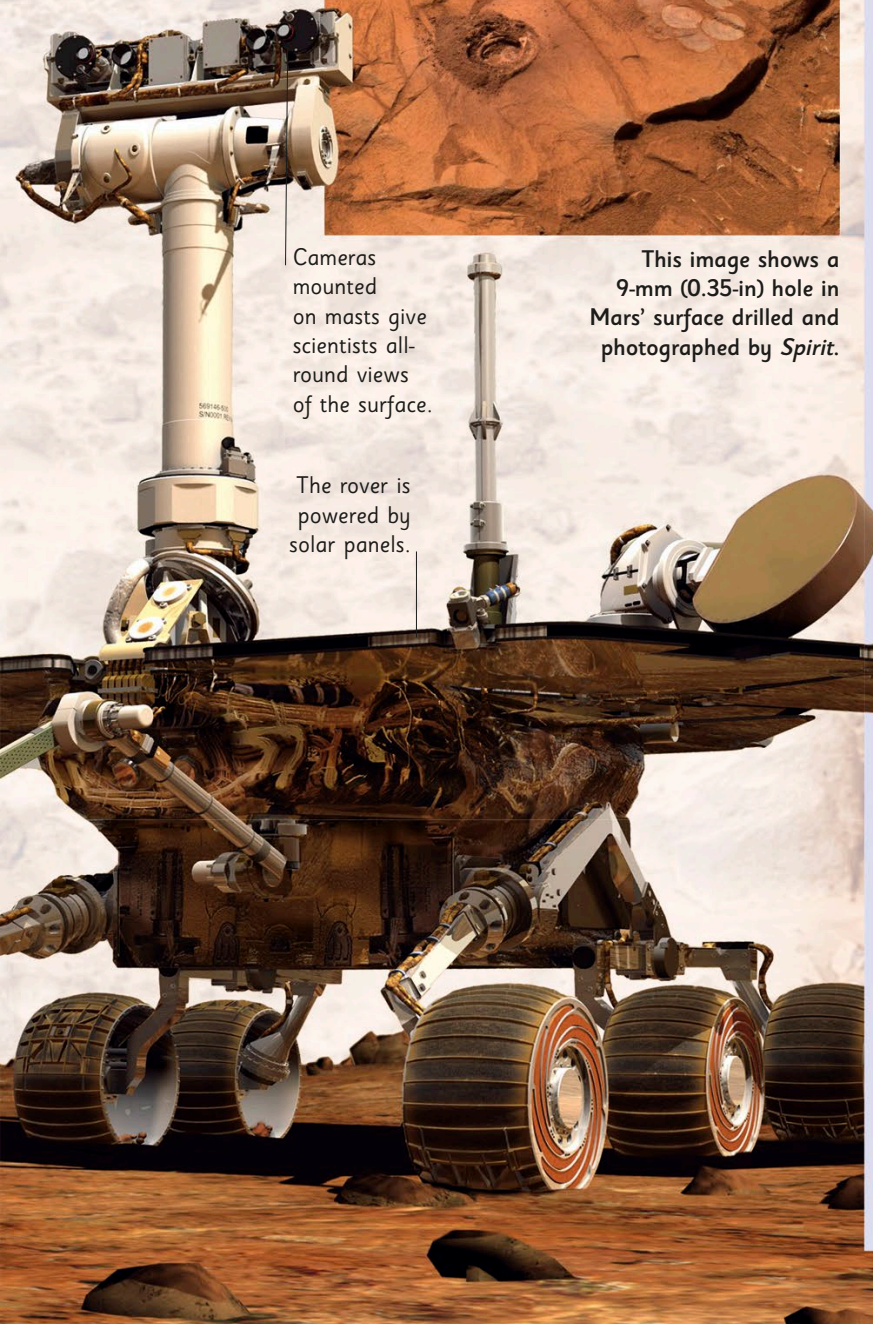
The NASA rover *Opportunity* holds the off-Earth roving record – by mid-2015 it completed 42.4 km (26.4 miles) on Mars.



Cameras mounted on masts give scientists all-round views of the surface.

This image shows a 9-mm (0.35-in) hole in Mars' surface drilled and photographed by *Spirit*.

The rover is powered by solar panels.



The future on Mars

Scientists are always searching for ways to unlock the secrets of the Red Planet. The suggested ideas include an aeroplane that could travel across its surface (above left) and a thermal probe that would penetrate its ice caps (above right).



In order to explore the potential of a colony in space, eight scientists lived in a self-contained dome, called Biosphere II, for two years during the early 1990s.

Living on Mars

If we do establish a base on Mars, it will have to be a self-contained structure that protects its inhabitants from both the atmosphere and the Sun's radiation. Below is an artist's impression of what a Martian base might look like.

